

E-Commerce Sales Analysis & Business Insights

Project Objective:

The primary objective of this project is to analyze e-commerce sales data and identify meaningful patterns and trends in business performance. The analysis focuses on key factors such as sales revenue, product performance, quantity sold, pricing, discounts, customer behavior, and sales trends over time.

The project aims to understand what is happening in the business, why certain sales patterns occur, what trends may continue in the future, and what actions can be taken to improve performance. Through this analysis, important factors influencing sales and revenue can be identified, along with areas that require attention or improvement.

The project applies the four fundamental types of data analysis — Descriptive, Diagnostic, Predictive, and Prescriptive Analysis — to transform raw e-commerce data into clear, actionable business insights that can support better decision-making, improve sales performance, and contribute to overall business growth.

Dataset Description:

The E-Commerce Sales Dataset contains transactional sales data covering customers, products, sales quantities, unit prices, discounts, and total sales amounts. The dataset is organized into different dimensions, such as Sales, Product, and Customer details. It can be used to analyze sales performance, product trends, customer behavior, discounts, and revenue patterns.

Column Description:

Column Name	Description
Month	Provides Month of the order given
Year	Provides the Year of the order given
Customer ID	Provides the Unique id given to each customer
Gender	Provides the Gender of the customer
Loyalty Level	Provides the Loyalty type of the customer
Product ID	Provides the Unique id given to each product
Category	Provides the Category of the Product
Sub Category	Provides the Sub Category of the product
Store ID	Provides the Unique id given to each Store
Region	Provides the Region of the store where it is located
City	Provides the City of the store where it is located
Store Type	Provides the Type of store
Quantity	Provides the Quantity of products sold
Unit Price	Provides the unit prices of each product
Discount	Provides the Discount given to Customer
Payment Type	Provides the Type of payment used
Revenue	Provides the Total amount or revenue produced for each Sale id

Data cleaning & transformation:

Formula used:

Converting the Raw Data into Table

The screenshot shows an Excel spreadsheet with a table of customer data. The table has 8 columns: Customer_ID, Name, Age, Gender, City, State, Country, and Loyalty_Level. The data is as follows:

Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	sheena bonilla	55	Male	Donaltdon	North Carolina	USA	
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	jeremy harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	paul fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Molnabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze
CUST1026	Amber Carlson	18	Female	East Michellemouth	Missouri	USA	Gold
CUST1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA	
CUST1028	Joshua Ryan	27	Female	New David	New Jersey	USA	Platinum

The screenshot shows the same Excel spreadsheet as above, but with a 'Find and Replace' dialog box open. The dialog box is set to find 'CUST' and replace it with 'CUST'. The 'Find what' field contains 'CUST' and the 'Replace with' field contains 'CUST'. The 'Find Next' button is highlighted.

=PROPER(TRIM(CLEAN([@Name])))

The screenshot shows an Excel spreadsheet with the following data:

Customer_ID	Name	Column1	Ag	Gende	City	State	Country	Loyalty_Level
CUST1000	sheena bonilla	=PROPER(TRIM(CLEAN([@Name])))			Donaldton	North Carolina	USA	
CUST1001	Eugene Nelson		26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	dr. Benjamin Price		29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	jeremy harris		59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins		64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn		24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson		29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd		29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard		53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach		64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page		53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez		28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert		55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez		32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen		53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez		49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott		24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz		60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss		43	Male	South William	Kansas	USA	
CUST1019	Michael Torres		46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen		59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez		36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	paul fox		33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele		52	Female	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins		40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor		59	Female	Vasquestown	California	USA	Bronze
CUST1026	Amber Carlson		18	Female	East Michellemouth	Missouri	USA	Gold
CUST1027	Leslie Johnson		31	Male	Mollyshire	Utah	USA	
CUST1028	Joshua Ryan		27	Female	New David	New Jersey	USA	Platinum

=IF(OR(ISBLANK([@Loyalty_Level]),[@Loyalty_Level]=0),"Unknown",[@Loyalty_Level])

The screenshot shows the same Excel spreadsheet, but with the formula bar updated to: =IF(OR(ISBLANK([@Loyalty_Level]),[@Loyalty_Level]=0),"Unknown",[@Loyalty_Level]). The Loyalty_Level column now contains the values "Unknown" for rows where the original value was blank or zero.

Customer_ID	Name	Ag	Gende	City	State	Country	Column1	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	=IF(OR(ISBLANK([@Loyalty_Level]),[@Loyalty_Level]=0),"Unknown",[@Loyalty_Level])	Platinum
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA		Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA		Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA		Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA		Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA		Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA		Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA		Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA		Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA		Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA		Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA		Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA		Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA		Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA		Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA		Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA		Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA		Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA		Unknown
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA		Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA		Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA		Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA		Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA		Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA		Platinum
CUST1025	Diana Taylor	59	Female	Vasquestown	California	USA		Bronze
CUST1026	Amber Carlson	18	Female	East Michellemouth	Missouri	USA		Gold
CUST1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA		Unknown
CUST1028	Joshua Ryan	27	Female	New David	New Jersey	USA		Platinum

Handling Duplicate Values

The screenshot shows an Excel spreadsheet with a table of customer data. A 'Remove Duplicates' dialog box is open, allowing the user to select which columns to check for duplicates. The columns selected are Customer_ID, Name, Age, Gender, and City. The 'My data has headers' checkbox is checked. The background table has 30 rows of data, including columns for Customer_ID, Name, Age, Gender, City, State, Country, and Loyalty_Level.

Finding the missing values using formulas

$$\text{unit price} = (\text{Tot amt} / (\text{qty} * (1 - \text{Discount})))$$

$$\text{Tot Amt} = \text{Qty} * \text{Unit Price} * (1 - \text{Discount})$$

$$\text{Grs Amt} = \text{Qty} * \text{Unit Price}$$

$$\text{Discount \%} = (\text{Grs Amt} - \text{Total Amt}) \div \text{Grs Amt} \times 100$$

OR

We can take the average of the individual product id's
 For example, the average unit price of PROD60 is \$398.886
 For Qty we can take the average of the Total Qty Product wise using pivot table and xlookup
 Discount values is replaced with Zero

=IF(OR(ISBLANK([@Quantity]),[@Quantity]=0),AVERAGE([Quantity]),[@Quantity])

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SUM =IF(OR(ISBLANK([@Quantity]),[@Quantity]=0),AVERAGE([Quantity]),[@Quantity])

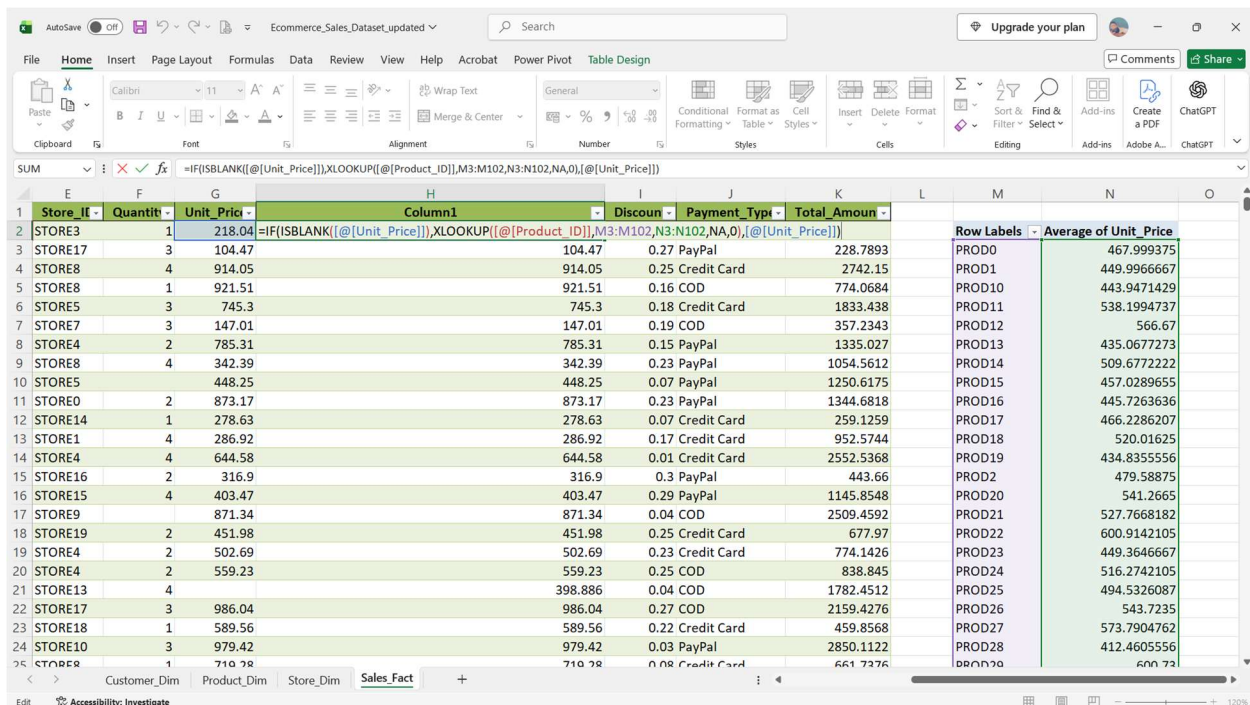
	B	C	D	E	F	G	
1	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Column1	Ur
2	07-07-2024	CUST1409	PROD97	STORE3	1	=IF(OR(ISBLANK([@Quantity]),[@Quantity]=0),AVERAGE([Quantity]),[@Quantity])	\$2
3	25-12-2023	CUST1467	PROD74	STORE17	3	3	\$1
4	03-06-2024	CUST1056	PROD49	STORE8	4	4	\$9
5	29-11-2023	CUST1346	PROD76	STORE8	1	1	\$9
6	06-01-2024	CUST1493	PROD59	STORE5	3	3	\$7
7	26-07-2024	CUST1116	PROD15	STORE7	3	3	\$1
8	12-07-2025	CUST1016	PROD56	STORE4	2	2	\$7
9	18-11-2023	CUST1152	PROD70	STORE8	4	4	\$3
10	04-05-2025	CUST1187	PROD99	STORE5	2	2	\$4
11	31-07-2025	CUST1464	PROD33	STORE0	2	2	\$8
12	13-02-2024	CUST1065	PROD24	STORE14	1	1	\$2
13	24-05-2025	CUST1320	PROD39	STORE1	4	4	\$2
14	24-08-2024	CUST1237	PROD88	STORE4	4	4	\$6
15	06-08-2025	CUST1443	PROD34	STORE16	2	2	\$3
16	21-06-2025	CUST1348	PROD24	STORE15	4	4	\$4
17	10-12-2024	CUST1435	PROD36	STORE9	2	2	\$8
18	13-09-2025	CUST1276	PROD65	STORE19	2	2	\$4
19	11-12-2024	CUST1079	PROD27	STORE4	2	2	\$5
20	13-08-2024	CUST1094	PROD39	STORE4	2	2	\$5
21	18-03-2024	CUST1239	PROD60	STORE13	4	4	
22	05-07-2025	CUST1144	PROD72	STORE17	3	3	\$9
23	04-06-2024	CUST1396	PROD43	STORE18	1	1	\$5

< > Solution Customer_Dim Product_Dim Store_Dim Sales_Fact + 130%

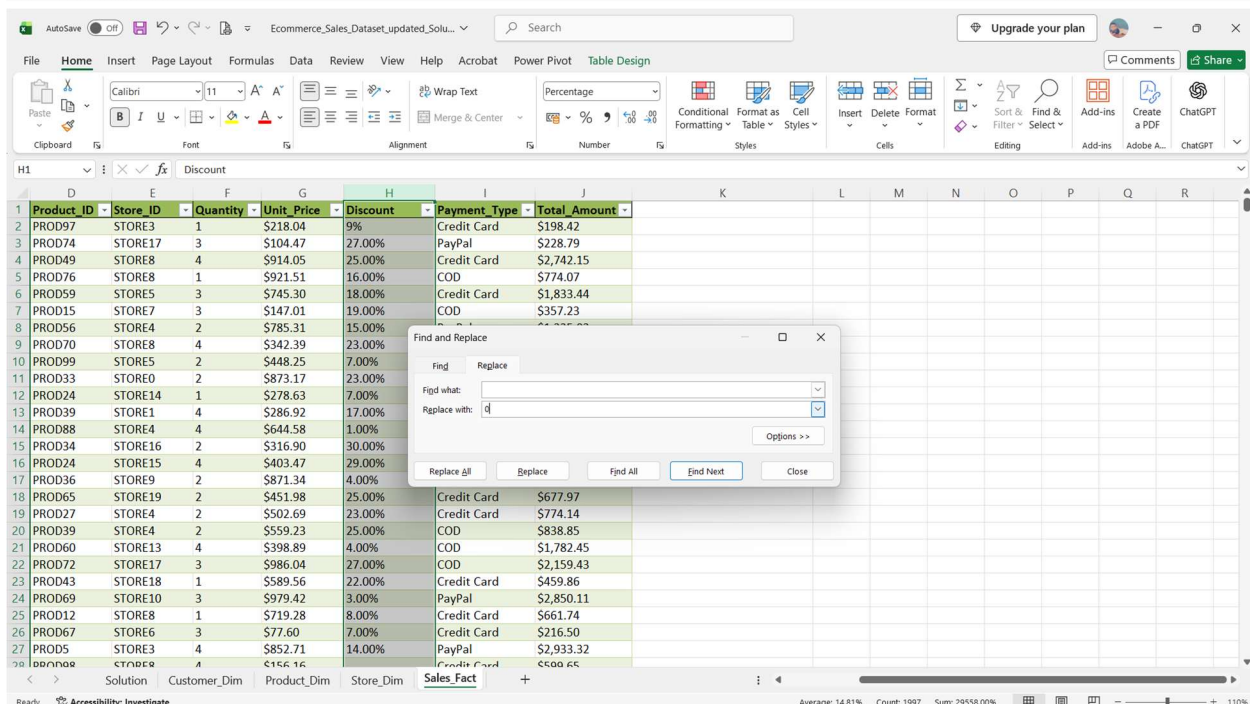
Edit Accessibility: Investigate

Using Pivot create a table to find the average unit price of all the product id's, and use Xlookup to find the missing unit price.

=IF(ISBLANK([@[Unit_Price]]),XLOOKUP([@[Product_ID]],M3:M102,N3:N102,NA,0),[@[Unit_Price]])



Store_ID	Quantity	Unit_Price	Column1	Discount	Payment_Type	Total_Amount
STORE3	1	218.04				
STORE17	3	104.47				
STORE8	4	914.05				
STORE8	1	921.51				
STORE5	3	745.3				
STORE7	3	147.01				
STORE4	2	785.31				
STORE8	4	342.39				
STORE5	4	448.25				
STORE0	2	873.17				
STORE14	1	278.63				
STORE1	4	286.92				
STORE4	4	644.58				
STORE16	2	316.9				
STORE15	4	403.47				
STORE9	2	871.34				
STORE19	2	451.98				
STORE4	2	502.69				
STORE4	2	559.23				
STORE13	4	398.886				
STORE17	3	986.04				
STORE18	1	589.56				
STORE10	3	979.42				
STORE8	1	710.78				



Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
PROD97	STORE3	1	\$218.04	9%	Credit Card	\$198.42
PROD74	STORE17	3	\$104.47	27.00%	PayPal	\$228.79
PROD49	STORE8	4	\$914.05	25.00%	Credit Card	\$2,742.15
PROD76	STORE8	1	\$921.51	16.00%	COD	\$774.07
PROD59	STORE5	3	\$745.30	18.00%	Credit Card	\$1,833.44
PROD15	STORE7	3	\$147.01	19.00%	COD	\$357.23
PROD56	STORE4	2	\$785.31	15.00%		
PROD70	STORE8	4	\$342.39	23.00%		
PROD99	STORE5	2	\$448.25	7.00%		
PROD33	STORE0	2	\$873.17	23.00%		
PROD24	STORE14	1	\$278.63	7.00%		
PROD39	STORE1	4	\$286.92	17.00%		
PROD88	STORE4	4	\$644.58	1.00%		
PROD34	STORE16	2	\$316.90	30.00%		
PROD24	STORE15	4	\$403.47	29.00%		
PROD36	STORE9	2	\$871.34	4.00%		
PROD65	STORE19	2	\$451.98	25.00%	Credit Card	\$677.97
PROD27	STORE4	2	\$502.69	23.00%	Credit Card	\$774.14
PROD39	STORE4	2	\$559.23	25.00%	COD	\$838.85
PROD60	STORE13	4	\$398.89	4.00%	COD	\$1,782.45
PROD72	STORE17	3	\$986.04	27.00%	COD	\$2,159.43
PROD43	STORE18	1	\$589.56	22.00%	Credit Card	\$459.86
PROD69	STORE10	3	\$979.42	3.00%	PayPal	\$2,850.11
PROD12	STORE8	1	\$719.28	8.00%	Credit Card	\$661.74
PROD67	STORE6	3	\$77.60	7.00%	Credit Card	\$216.50
PROD5	STORE3	4	\$852.71	14.00%	PayPal	\$2,933.32
PROD98	STORE8	4	\$166.16		Credit Card	\$661.74

Using a formula to calculate the Total Amount

$$=[@Quantity]*[@Unit_Price]*(1-[@Discount])$$

The screenshot shows an Excel spreadsheet with a table of sales data. The formula bar at the top displays the formula
$$=[@Quantity]*[@Unit_Price]*(1-[@Discount])$$
 for the Total Amount column.

Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
PROD97	STORE3	1	\$218.04	9%	Credit Card	=\$[Quantity]*[Unit_Price]*(1-[Discount])
PROD74	STORE17	3	\$104.47	27.00%	PayPal	\$228.79
PROD49	STORE8	4	\$914.05	25.00%	Credit Card	\$2,742.15
PROD76	STORE8	1	\$921.51	16.00%	COD	\$774.07
PROD59	STORE5	3	\$745.30	18.00%	Credit Card	\$1,833.44
PROD15	STORE7	3	\$147.01	19.00%	COD	\$357.23
PROD56	STORE4	2	\$785.31	15.00%	PayPal	\$1,335.03
PROD70	STORE8	4	\$342.39	23.00%	PayPal	\$1,054.56
PROD99	STORE5	2	\$448.25	7.00%	PayPal	\$1,033.93
PROD33	STORE0	2	\$873.17	23.00%	PayPal	\$1,344.68
PROD24	STORE14	1	\$278.63	7.00%	Credit Card	\$259.13
PROD39	STORE1	4	\$286.92	17.00%	Credit Card	\$952.57
PROD88	STORE4	4	\$644.58	1.00%	Credit Card	\$2,552.54
PROD34	STORE16	2	\$316.90	30.00%	PayPal	\$443.66
PROD24	STORE15	4	\$403.47	29.00%	PayPal	\$1,145.85
PROD36	STORE9	2	\$871.34	4.00%	COD	\$2,074.65
PROD65	STORE19	2	\$451.98	25.00%	Credit Card	\$677.97
PROD27	STORE4	2	\$502.69	23.00%	Credit Card	\$774.14
PROD39	STORE4	2	\$559.23	25.00%	COD	\$838.85
PROD60	STORE13	4	\$398.89	4.00%	COD	\$1,531.72
PROD72	STORE17	3	\$986.04	27.00%	COD	\$2,159.43
PROD43	STORE18	1	\$589.56	22.00%	Credit Card	\$459.86
PROD69	STORE10	3	\$979.42	3.00%	PayPal	\$2,850.11
PROD12	STORE8	1	\$719.28	8.00%	Credit Card	\$661.74
PROD67	STORE6	3	\$77.60	7.00%	Credit Card	\$216.50
PROD5	STORE3	4	\$852.71	14.00%	PayPal	\$2,933.32
PROD13	STORE4	4	\$166.16	0.00%	Credit Card	\$664.64

Used Text function to split the date into Month & Year

$$=TEXT([@Order_Date], "MMM YYYY")$$

The screenshot shows an Excel spreadsheet with a table of sales data. The formula bar at the top displays the formula
$$=TEXT([@Order_Date], "MMM YYYY")$$
 for the Order Date column.

Order_Date	Order Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
07-07-2024	=TEXT([@Order_Date], "MMM YYYY")		PROD97	STORE3	1	\$218.04	9%	Credit Card	\$198.42
25-12-2023	Dec 2023	CUST1467	PROD74	STORE17	3	\$104.47	27.00%	PayPal	\$228.79
03-06-2024	Jun 2024	CUST1056	PROD49	STORE8	4	\$914.05	25.00%	Credit Card	\$2,742.15
29-11-2023	Nov 2023	CUST1346	PROD76	STORE8	1	\$921.51	16.00%	COD	\$774.07
06-01-2024	Jan 2024	CUST1493	PROD59	STORE5	3	\$745.30	18.00%	Credit Card	\$1,833.44
26-07-2024	Jul 2024	CUST1116	PROD15	STORE7	3	\$147.01	19.00%	COD	\$357.23
12-07-2025	Jul 2025	CUST1016	PROD56	STORE4	2	\$785.31	15.00%	PayPal	\$1,335.03
18-11-2023	Nov 2023	CUST1152	PROD70	STORE8	4	\$342.39	23.00%	PayPal	\$1,054.56
04-05-2025	May 2025	CUST1187	PROD99	STORE5	2	\$448.25	7.00%	PayPal	\$1,033.93
31-07-2025	Jul 2025	CUST1464	PROD33	STORE0	2	\$873.17	23.00%	PayPal	\$1,344.68
13-02-2024	Feb 2024	CUST1065	PROD24	STORE14	1	\$278.63	7.00%	Credit Card	\$259.13
24-05-2025	May 2025	CUST1320	PROD39	STORE1	4	\$286.92	17.00%	Credit Card	\$952.57
24-08-2024	Aug 2024	CUST1237	PROD88	STORE4	4	\$644.58	1.00%	Credit Card	\$2,552.54
06-08-2025	Aug 2025	CUST1443	PROD34	STORE16	2	\$316.90	30.00%	PayPal	\$443.66
21-06-2025	Jun 2025	CUST1348	PROD24	STORE15	4	\$403.47	29.00%	PayPal	\$1,145.85
10-12-2024	Dec 2024	CUST1435	PROD36	STORE9	2	\$871.34	4.00%	COD	\$2,074.65
13-09-2025	Sep 2025	CUST1276	PROD65	STORE19	2	\$451.98	25.00%	Credit Card	\$677.97
11-12-2024	Dec 2024	CUST1079	PROD27	STORE4	2	\$502.69	23.00%	Credit Card	\$774.14
13-08-2024	Aug 2024	CUST1094	PROD39	STORE4	2	\$559.23	25.00%	COD	\$838.85
18-03-2024	Mar 2024	CUST1239	PROD60	STORE13	4	\$398.89	4.00%	COD	\$1,531.72
05-07-2025	Jul 2025	CUST1144	PROD72	STORE17	3	\$986.04	27.00%	COD	\$2,159.43
04-06-2024	Jun 2024	CUST1396	PROD43	STORE18	1	\$589.56	22.00%	Credit Card	\$459.86
12-04-2024	Apr 2024	CUST1476	PROD69	STORE10	3	\$979.42	3.00%	PayPal	\$2,850.11
07-07-2024	Jul 2024	CUST1368	PROD5	STORE3	4	\$852.71	14.00%	PayPal	\$2,933.32
07-07-2024	Jul 2024	CUST1368	PROD13	STORE4	4	\$166.16	0.00%	Credit Card	\$664.64

Used Vlookup to combine sheets using Cust ID, Prod ID, Store ID into the Sales_Sheet

=VLOOKUP([@[Product_ID]],Product_Dim!A:C,3,0)

Order Date	Month	Customer ID	Gender	Loyalty Level	Product ID	Category	Sub-Cat	Store ID	Region	City	Store Ty	Quantity	Unit Price	Discount	Payment Type	Total Amount
07-07-2024	Jul 2024	CUST1409	Male	Gold	PROD97	Sports	Team Spor	STORE3	North	Hoodberg	Hoodberg	1	\$218.04	9%	Credit Card	\$198.42
25-12-2023	Dec 2023	CUST1467	Female	Gold	PROD74	Electronic	Audio	STORE17	North	Bensonhah	Bensonhave	3	\$104.47	27.00%	PayPal	\$228.79
03-06-2024	Jun 2024	CUST1056	Female	Gold	PROD49	Beauty	Fragrance	STORE8	West	Thompson	Thompsonb	4	\$914.05	25.00%	Credit Card	\$2,742.15
29-11-2023	Nov 2023	CUST1346	Female	Platinum	PROD76	Electronic	Computer	STORE8	West	Thompson	Thompsonb	1	\$921.51	16.00%	COD	\$774.07
06-01-2024	Jan 2024	CUST1493	Male	Silver	PROD59	Beauty	Haircare	STORE5	North	New Mich	New Michae	3	\$745.30	18.00%	Credit Card	\$1,833.44
26-07-2024	Jul 2024	CUST1116	Female	Platinum	PROD15	Electronic	Appliances	STORE7	South	Williamsfu	Williamsfur	3	\$147.01	19.00%	COD	\$357.23
12-07-2025	Jul 2025	CUST1016	Female	Bronze	PROD56	Home	Decor	STORE4	South	Port Sharo	Port Sharon	2	\$785.31	15.00%	PayPal	\$1,335.03
18-11-2023	Nov 2023	CUST1152	Female	Platinum	PROD70	Home	Storage	STORE8	West	Thompson	Thompsonb	4	\$342.39	23.00%	PayPal	\$1,054.56
04-05-2025	May 2025	CUST1187	Female	Platinum	PROD99	Home	Kitchen	STORE5	North	New Mich	New Michae	2	\$448.25	7.00%	PayPal	\$1,033.93
31-07-2025	Jul 2025	CUST1464	Female	Platinum	PROD33	Home	Decor	STORE0	East	Washingtc	Washington	2	\$873.17	23.00%	PayPal	\$1,344.68
13-02-2024	Feb 2024	CUST1065	Female	Silver	PROD24	Sports	Outdoor	STORE14	South	Christophe	Christopher	1	\$278.63	7.00%	Credit Card	\$259.13
24-05-2025	May 2025	CUST1320	Female	Bronze	PROD39	Electronic	Mobile	STORE1	North	Danielmo	Danielmout	4	\$286.92	17.00%	Credit Card	\$952.57
24-08-2024	Aug 2024	CUST1237	Female	Gold	PROD88	Sports	Accessory	STORE4	South	Port Sharo	Port Sharon	4	\$644.58	1.00%	Credit Card	\$2,552.54
06-08-2025	Aug 2025	CUST1443	Male	Silver	PROD34	Electronic	Computer	STORE16	West	New Jimm	New Jimmyr	2	\$316.90	30.00%	PayPal	\$443.66
21-06-2025	Jun 2025	CUST1348	Male	Silver	PROD24	Sports	Outdoor	STORE15	North	East Jenni	East Jennife	4	\$403.47	29.00%	PayPal	\$1,145.85
10-12-2024	Dec 2024	CUST1435	Female	Silver	PROD36	Clothing	Tops	STORE9	South	East Johns	East Johnsta	2	\$871.34	4.00%	COD	\$2,074.65
13-09-2025	Sep 2025	CUST1276	Male	Gold	PROD65	Sports	Fitness	STORE19	East	Vasquezbc	Vasquezbor	2	\$451.98	25.00%	Credit Card	\$677.97
11-12-2024	Dec 2024	CUST1079	Female	Platinum	PROD27	Home	Decor	STORE4	South	Port Sharo	Port Sharon	2	\$502.69	23.00%	Credit Card	\$774.14
13-08-2024	Aug 2024	CUST1094	Female	Gold	PROD39	Electronic	Mobile	STORE4	South	Port Sharo	Port Sharon	2	\$559.23	25.00%	COD	\$838.85
18-03-2024	Mar 2024	CUST1239	Female	Platinum	PROD60	Beauty	Fragrance	STORE13	East	Johnsonvi	Johnsonvil	4	\$398.89	4.00%	COD	\$1,531.72
05-07-2025	Jul 2025	CUST1144	Female	Bronze	PROD72	Sports	Fitness	STORE17	North	Bensonhah	Bensonhave	3	\$986.04	27.00%	COD	\$2,159.43
04-06-2024	Jun 2024	CUST1396	Female	Silver	PROD43	Electronic	Computer	STORE18	East	East Steve	East Stevech	1	\$589.56	22.00%	Credit Card	\$459.86
12-04-2024	Apr 2024	CUST1476	Female	Gold	PROD69	Clothing	Bottoms	STORE10	East	East David	East Davidvi	3	\$979.42	3.00%	PayPal	\$2,850.11
23-06-2024	Jun 2024	CUST1368	Female	Platinum	PROD12	Beauty	Haircare	STORE8	West	Thompson	Thompsonb	1	\$719.28	8.00%	Credit Card	\$661.74
08-09-2025	Sep 2025	CUST1186	Male	Silver	PROD67	Electronic	Accessory	STORE6	East	Port Josep	Port Joseph	3	\$77.60	7.00%	Credit Card	\$216.50
17-09-2024	Sep 2024	CUST1381	Male	Silver	PROD5	Clothing	Accessory	STORE3	North	Hoodberg	Hoodberg	4	\$852.71	14.00%	PayPal	\$2,933.32
03-01-2024	Jan 2024	CUST1213	Female	Platinum	PROD98	Electronic	Audio	STORE8	West	Thompson	Thompsonb	4	\$156.16	0.00%	Credit Card	\$624.64
22-02-2025	Feb 2025	CUST1200	Male	Platinum	PROD77	Electronic	Appliances	STORE8	West	Thompson	Thompsonb	3	\$815.49	17.00%	Credit Card	\$2,030.57
23-09-2025	Sep 2025	CUST1148	Female	Bronze	PROD13	Electronic	Accessory	STORE2	South	North Nan	North Nancy	2	\$74.21	11.00%	PayPal	\$132.09

Used Descriptive Statistics using Toolpak

Store Type	Quantity	Unit Price	Discount	Payment Type	Revenue
Hoodberg	1	\$218.04	9%	Credit Card	\$198.42
Bensonhaven	3	\$104.47	27.00%	PayPal	\$228.79
Thompsonborough	4	\$914.05	25.00%	Credit Card	\$2,742.15
Thompsonborough	1	\$921.51	16.00%	COD	\$774.07
New Michaelchester	3	\$745.30	18.00%	Credit Card	\$1,833.44
Williamsfurt	3	\$147.01	19.00%	COD	\$357.23
Port Sharon	2	\$785.31	15.00%	PayPal	\$1,335.03
Thompsonborough	4	\$342.39	23.00%	PayPal	\$1,054.56
New Michaelchester	2	\$448.25	7.00%	PayPal	\$1,033.93
Washingtonfurt	2	\$873.17	23.00%	PayPal	\$1,344.68
Christopherland	1	\$278.63	7.00%	Credit Card	\$259.13
Danielmout	4	\$286.92	17.00%	Credit Card	\$952.57
Port Sharon	4	\$644.58	1.00%	Credit Card	\$2,552.54
New Jimmymouth	2	\$316.90	30.00%	PayPal	\$443.66
East Jenniferstad	4	\$403.47	29.00%	PayPal	\$1,145.85
East Johnstad	2	\$871.34	4.00%	COD	\$2,074.65
Vasquezborough	2	\$451.98	25.00%	Credit Card	\$677.97
Port Sharon	2	\$502.69	23.00%	Credit Card	\$774.14
Port Sharon	2	\$559.23	25.00%	COD	\$838.85
Johnsonville	4	\$398.89	4.00%	COD	\$1,531.72
Bensonhaven	3	\$986.04	27.00%	COD	\$2,159.43
East Stevechester	1	\$589.56	22.00%	Credit Card	\$459.86
East Davidville	3	\$979.42	3.00%	PayPal	\$2,850.11
Thompsonborough	1	\$719.28	8.00%	Credit Card	\$661.74
Port Josephberg	3	\$77.60	7.00%	Credit Card	\$216.50
Hoodberg	4	\$852.71	14.00%	PayPal	\$2,933.32
Thompsonborough	4	\$156.16	0.00%	Credit Card	\$624.64
Thompsonborough	3	\$815.49	17.00%	Credit Card	\$2,030.57
North Nancyview	2	\$74.21	11.00%	PayPal	\$132.09

Descriptive Statistics	
Revenue	
Mean	1087.41183
Standard Error	18.3124919
Median	871.40815
Mode	#N/A
Standard Deviation	818.959536
Sample Variance	670694.721
Kurtosis	0.05671365
Skewness	0.88084375
Range	3889.7838
Minimum	18.459
Maximum	3908.2428
Sum	2174823.66
Count	2000
Largest(1)	3908.2428
Smallest(1)	18.459
Confidence Level(95.0%)	35.9135695

Data Analysis

Analysis Tools

- Anova: Single Factor
- Anova: Two-Factor With Replication
- Anova: Two-Factor Without Replication
- Correlation
- Covariance
- Descriptive Statistics**
- Exponential Smoothing
- F-Test Two-Sample for Variances
- Fourier Analysis
- Histogram

OK

Cancel

Help

Visualization and Insights Arrived with screenshot of visualization Report/Dashboard screenshot:

- **Descriptive analysis:**
 - What is the total sales?
 - Which product category generated the highest sales?
 - Which region performed best?
 - Which Year had the highest sales?
- **Diagnostic Analysis:**
 - Why did the top category generate higher sales?
 - Why is the profit higher/lower for certain products?
- **Predictive Analysis:**
 - Based on the sales trend, what could next month's sales be?
 - Which category is likely to perform well in the future?
- **Prescriptive Analysis:**
 - Which category should the company focus on?
 - Should the company increase or reduce discounts?
 - What 3 actions should the company take to improve profit?

Dashboard Image:



Final 4 type of insights as storytelling:

We had 2,000 total orders placed by happy customers. Together, they bought 4,960 total items. Our total sales revenue came out to \$21,74,824, and the average order was about \$1,087. Out of all our product categories, Sports brought in the most money, closely followed by Home and Electronics. I noticed that Sports and Home did so well because both male and female customers bought them in steady, balanced quantities. When people paid for their items, most of them used Credit Cards and PayPal, which made transactions smooth and easy compared to cash-on-delivery. The trend lines show that our upcoming monthly sales will likely settle comfortably between \$10,000 and \$15,000 as regular seasonal cycles continue. Electronics and Sports look like they will keep growing and remain customer favorites. Put more advertising and stock into our top-selling categories like Sports and Home. Instead of giving giant percentage discounts that cut into our profits, we should try offering bundle deals. Our

West region only has 12% of sales, so we should run special promos there to catch up with the North and East.

Conclusion:

This project taught me that data isn't just a scary spreadsheet, it is actually a story about people shopping, buying, and helping a business grow. I am so proud of building my first dashboard and finding these insights.